No. of Printed Pages : 5

Sem-II-UG-CompSc-MjP-3(NEP)
(Data Structure)

07. 04/08/2025

2024-25

Time - 3 hours

Full Marks - 100

Answer all parts as per instructions.
Figures in the right hand margin indicate marks.

PART – I

1. Answer all questions by choosing the correct answer from the given alternatives. [1 × 10

(a) Which of the following operations access each record exactly once so that certain items may be processed ?

(i) Inserting (ii) Deleting

(iii) Traversing (iv) Searching

(b) Which of the following is a non-linear data structure ?

(i) Array (ii) Stack

(iii) Linked List (iv) Graph

(c) Each program module contains its own list of variables called __________.

(i) Global (ii) Local

(iii) Search (iv) Binary

P.T.O.

(g) Explain Inorder traversal in a binary tree with an example.

(h) Suppose a stack has elements 5, 10, 15 (top is 15). Show the stack content after one pop and one push of 20.

(i) If a binary search tree has nodes inserted in order 50, 30, 70, 20, 40, draw the tree structure.

PART – III

3. Answer any eight of the following questions within 250 words each. [5 × 8

(a) Write an algorithm and a C program to perform binary search in a sorted array.

(b) Write a program to implement a stack using a linked list. Perform push, pop and display operations.

(c) What are the limitations of arrays ? How does a linked list overcome these limitations ?

(d) Convert the infix expression (A + B)*(C – D) to postfix and explain the steps involved.

(e) Describe the structure and application of a priority queue.

(f) Write a snippet to insert a node at the beginning and the end of a singly linked list.

(g) What is an AVL tree ? Explain rotations used in AVL trees.

(h) What is a Binary Search Tree (BST) ? Write a function to insert nodes in a BST.

(i) Describe the BFS traversal algorithm for a graph and explain its working with an example.

(j) Define M-way search trees. How are they different from binary trees ?

PART – IV

*Answer **any four** questions within 800 words each.*

4. Write a program for Preorder and Postorder tree traversal. [8

5. Write a program in C to perform the following operations on a one-dimensional array : insertion, deletion and search. Explain each operation. [8

6. Explain the insertion and deletion operations in AVL trees with examples. How does it maintain balance ? [8

7. Explain different methods of traversing a binary tree. The inorder traversal is given as : "DBFEAGCLJHK". Find the preorder traversal of the tree. [8

8. Write a C program to implement a singly linked list and perform all the following operations : insert at beginning, insert at end, delete a node and traverse. [8

(d) The term ________ will be used for an independent algorithmic module which solves a particular problem.

(i) Program (ii) Logic

(iii) Name (iv) Procedure

(e) Which of the following items are not part of the array declaration ?

(i) Name of the array (ii) Data type of the array

(iii) Index of the array (iv) Length of the array

(f) ________ refers to the situation where one wants to delete data from a data structure that is empty.

(i) Free Storage (ii) Underflow

(iii) Overflow (iv) Compaction

(g) Which of the following is not an application of a stack ?

(i) Recursion (ii) Backtracking

(iii) Tower of Hanoi (iv) Priority Scheduling

(h) In a ________, the left and right subtrees for any given node differ no more than one.

(i) Balanced binary tree

(ii) Weight-balanced binary tree

(iii) Height-balanced binary tree

(iv) Binary search tree

(i) In ________, the problem of sorting a set is reduced to the problem of sorting two smaller sets.

(i) Quick Sort (ii) Heap Sort

(iii) Bubble Sort (iv) Merge Sort

(j) ________ is the insert operation of the queue structure.

(i) Dequeue (ii) Enqueue

(iii) Queue (iv) Linear list

PART – II

2. Answer all questions within 50 words each. [2 × 9

(a) What is the difference between Linear Search and Binary Search ?

(b) List the differences between Stack and Queue.

(c) Write a C code fragment to check if a stack is full using an array of size 100.

(d) Define Circular Queue. What problem does it solve compared to a linear queue ?

(e) Write a C function to perform PUSH operation in a stack implemented using an array.

(f) List two advantages and two disadvantages of dynamic memory allocation.